Day 38 EBIKE101 Student Print Packet

CTS Cohort 6 | EBIKE101 | Tuesday 2026-09-08

Day 38 EBIKE101 Morning Quiz — Student Copy

Program: CTS Cohort 6

Course: EBIKE101

Date: Tuesday, 2026-09-08

Topic: E-bike Homunculous diagnosis and finish-work planning

Name: _____

Directions

Circle the best answer for each question. This quiz is about safe diagnosis, not guessing bravely at wires like a tiny electrician raccoon.

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1. Before diagnosing an e-bike electrical problem, what should happen first?

A. Connect every loose wire and see what turns on

B. Confirm the system is safe to inspect and de-energized when appropriate

C. Spin the motor by hand as fast as possible

D. Replace the controller immediately

2. What is the best first step when looking at an unfinished e-bike build?

A. Make a full system inventory and note what is present, missing, loose, or unknown

B. Assume the last person wired it correctly

C. Start cutting zip ties so every wire is free

D. Connect the battery and test under load

3. What is the job of a fuse or breaker in an e-bike power circuit?

A. Make the motor run faster

B. Stop unsafe current before wiring or parts are damaged

C. Increase battery voltage

D. Turn pedal assist on automatically

4. If the display does not power on, which item is a reasonable early check?

A. Tire tread pattern

B. Battery voltage, main switch, fuse, and display/controller connector

C. Seat height only

D. Rear reflector angle

5. What does a missing throttle usually mean during diagnosis?

A. The whole project is impossible

B. The team should document the missing control input and verify whether the system can operate by PAS/display without it

C. The battery must be bad

D. The motor wires should be reversed

6. Why should wire color not be trusted by itself?

A. Wire color is always random

B. Wire color is a clue, but repairs, adapters, and different manufacturers can change meanings

C. Only red wires carry electricity

D. Black wires are never connected to anything

7. What is the safest way to use a continuity test?

A. On a de-energized circuit or component, following meter instructions

B. Across a live battery pack

C. Across the controller's main power leads while the battery is connected

D. Anywhere, because continuity mode is harmless

8. What can happen if an e-bike brake cutoff sensor is active or misadjusted?

A. The controller may prevent motor output

B. The battery voltage will double

C. The tire pressure will drop

D. The chain will automatically tighten

9. What does PAS stand for in most e-bike systems?

A. Pedal Assist Sensor/System

B. Power Amplifier Switch

C. Positive Axle Support

D. Panel Alignment Screw

10. Why does motor cable strain relief matter?

A. It makes the bike look expensive

B. It prevents cable damage from pulling, rubbing, bending, or wheel movement

C. It increases top speed

D. It changes the tire size

11. If a fuse is blown, what should the team do before simply replacing it?

A. Bypass it with wire

B. Ask what caused the fuse to blow and inspect for shorts, wrong connections, or overload

C. Install the largest fuse available

D. Tape over the fuse holder

12. What does it mean to trace a circuit?

A. Draw a pretty version of the bike after it works

B. Follow the actual path from source, through protection/control, to the load

C. Guess which wire is correct based on color

D. Skip testing because the labels look professional

13. Which mechanical item is critical before powered wheel testing?

A. Handlebar grip color

B. Axle nuts/torque arms, wheel security, brake function, and clear chain/wheel movement

C. Sticker placement

D. Bell volume

14. What is a good practice for first powered motor testing in class?

A. Full throttle with someone holding the rear wheel

B. Secure the bike, keep people clear of moving parts, use low power, and have instructor control

C. Ride it through the shop immediately

D. Test with no brakes installed

15. What should be recorded in a diagnostic work log?

A. Only the final answer

B. Symptoms, observations, measurements, changes made, parts needed, and next step

C. Only who touched the bike last

D. Nothing if the team feels confident

16. What does a controller usually do in an e-bike system?

A. Converts rider/display inputs into controlled power sent to the motor

B. Inflates the tires

C. Replaces the chain

D. Makes the battery physically larger

17. What should happen if a component smells burnt, gets unusually hot, or shows melted wiring?

A. Keep testing to gather more data

B. Stop, disconnect power if safe, and get instructor review

C. Spray it with water

D. Wrap it in tape and continue

18. Why is connector inspection important?

A. Small pins, bent terminals, loose plugs, corrosion, or mismatched connectors can stop a system from working

B. Connectors are only cosmetic

C. Connectors never fail

D. Every connector can plug into every other connector safely

19. What is the difference between a symptom and a root cause?

A. They are always the same thing

B. A symptom is what you observe; a root cause is the underlying reason it is happening

C. A root cause is always the most expensive part

D. A symptom only matters if the motor spins

20. When is the Homunculous closer to “finished”?

A. When one person says “looks good”

B. When mechanical safety, electrical safety, control function, wiring support, needed parts, and documentation are verified

C. When every wire is hidden

D. When the fastest student rides it first

Exit check

Write one thing your team must verify before this e-bike should be powered or ridden:

_____

_____

Day 38 EBIKE101 Diagnostic Worksheet

Program: CTS Cohort 6

Course: EBIKE101

Date: Tuesday, 2026-09-08

Project: E-bike Homunculous

Team members: _____

Purpose

Use this worksheet to diagnose what the Homunculous needs before it can be safely finished, powered, and eventually ridden.

Safety boundary

Do not connect the battery, bypass protection, open battery cases, test live battery circuits in continuity mode, spin the motor under power, or ride the bike without instructor approval.

1. Whole-bike inventory

2. System trace

Fill this out before powered testing.

3. Symptom list

4. Missing parts / supplies list

5. Safe work approved today

Only list work the instructor has approved.

6. Evidence log

7. Exit reflection

What is one thing we verified today?

_____

What is one thing we still do not know?

_____

What is the next safest work step?

_____

Day 38 EBIKE101 Afternoon Lab — Homunculous Finish-Work Diagnosis

Program: CTS Cohort 6

Course: EBIKE101

Date: Tuesday, 2026-09-08

Lab type: Live project diagnosis and approved finish work

Lab objective

Students will diagnose the current state of the E-bike Homunculous, identify what is required to finish it, and begin only the safe work that has been verified and approved.

Instructor setup

Before students begin:

1. Place the bike where all sides can be accessed.
2. Keep battery connection under instructor control.
3. Remove or isolate any obviously unsafe condition.
4. Stage hand tools, meter, labels, zip ties, and worksheet.
5. Define the boundary: diagnosis first, power later only if earned by evidence.

Lab roles

Assign or rotate:

- Mechanical lead: checks frame, wheels, brakes, fasteners, chain path.
- Electrical tracer: maps source, protection, controller, controls, motor.
- Documentation lead: worksheet, board, photo evidence, parts list.
- Safety observer: watches pinch points, battery boundary, tool use, moving parts.
- Parts lead: organizes missing parts, hardware, adapters, labels.

Part 1 — Mechanical readiness inspection

Students inspect:

- Wheel security
- Axle nut condition
- Torque arm or anti-rotation support if applicable
- Brake function
- Tire condition and pressure
- Chain alignment
- Crank/pedal condition
- Sharp edges
- Cable rub points
- Loose fasteners
- Battery/controller mounting

Stop if:

- Motor wheel can shift in dropout

- Brakes cannot stop the wheel
- Chain path is unsafe
- Battery mount is loose
- Any wire can reach tire, chain, brake rotor, or crank

Part 2 — Electrical/control inventory

Students identify and document:

- Battery voltage/rating if labeled
- Controller label/rating if visible
- Main power connector
- Fuse/protection location
- Display connector
- PAS connector/sensor/magnet ring
- Throttle presence or absence
- Brake cutoff connectors
- Motor cable
- Accessory wires
- Damaged connectors
- Unknown wires

Rule:

Unknown wires stay unknown until traced, labeled, or verified. Guessing does not become truth just because someone says it confidently in a shop voice.

Part 3 — Diagnostic decision tree

Use this order:

1. Is the bike mechanically safe to inspect?
2. Is the battery appropriate and controlled?
3. Is there protection between battery and controller?
4. Does the controller have a secure mount?
5. Are control inputs present?
6. Are brake cutoffs present and not stuck active?
7. Is the PAS sensor installed and aligned?
8. Is the throttle present, missing, or optional for this controller?
9. Is the motor cable undamaged and supported?
10. What must be fixed before first power-up?

Part 4 — Approved finish work

Allowed student work may include:

- Brake adjustment
- Tightening non-critical hardware under supervision

- Labeling wires
- Improving cable routing
- Adding strain relief
- Cleaning mounting surfaces
- Organizing connectors
- Creating a parts-needed list
- Documenting evidence
- Non-powered meter checks where appropriate

Instructor-only or instructor-controlled work:

- Battery connection
- First power-up
- Controller testing
- Motor spin testing
- Fuse replacement if cause is unknown
- Any modification to high-current wiring
- Any ride test

Part 5 — First power-up gate

Do not perform a powered test unless all are checked:

- Mechanical assembly secure
- Brakes functional
- Drive wheel controlled
- Battery voltage/rating appropriate
- Fuse/protection path verified
- Controller mounted
- No melted/damaged wires
- No unrestrained motor cables
- No hands near moving parts
- Instructor gives explicit approval

Completion target

The lab is complete when the team can answer:

1. What works or appears complete?
2. What is missing?
3. What is unsafe?
4. What is unknown?
5. What should be done next?

Finishing the bike is excellent. Finishing the diagnosis is required.